$H_0: p \le 0.25$ $H_A: p > 0.25$

Discussion Section Worksheet 7

right.

1. The Isla Vista Water District claims that more than 25% of homeowners have high efficiency appliances. A company representative selects a random sample of 138 homeowners to participate in a survey. Assume that they are testing the claim at a 10% significance level.

   a) Define the parameter of interest.

   population proportion p. percentage of

   b) Define the random variable of interest.

   sample proportion $\hat{p}$. percentage of from

   c) Determine the distribution of the test statistic. (Show any conditions you have checked.)

   $z = \dfrac{\hat{p} - p = E(\hat{p})}{\sqrt{\dfrac{p(1-p)}{n}} = sd(\hat{p})}$ z: approximate standard normal

   if $np = 138 \times 25\% = 34.5 > 10$.
   $n(1-p) = 138 \times 75\% > 10$.

   d) The hypotheses to test the claim are: $H_0 : p \le 0.25$ $H_A : p > 0.25$ (claim)

   From the survey responses, they counted the number of homeowners who have high efficiency appliances (out of 138) and found the calculated p-value to be significant. Explain what this means, and whether or not you reject the null hypothesis.

   $X \sim Bin(n, p)$

   $\hat{p} \overset{approx}{\sim} N\left(p, \sqrt{\dfrac{p(1-p)}{n}}\right)$

   if: $\begin{cases} np \ge 10 \\ n(1-p) \ge 10 \end{cases}$ only for proportion

   p-value $= P(|Z| > |z|) < \alpha \Leftrightarrow p$: significant.

   reject H_0.

   e) State your conclusion. Please use sentence(s).

   At 5% significance level.

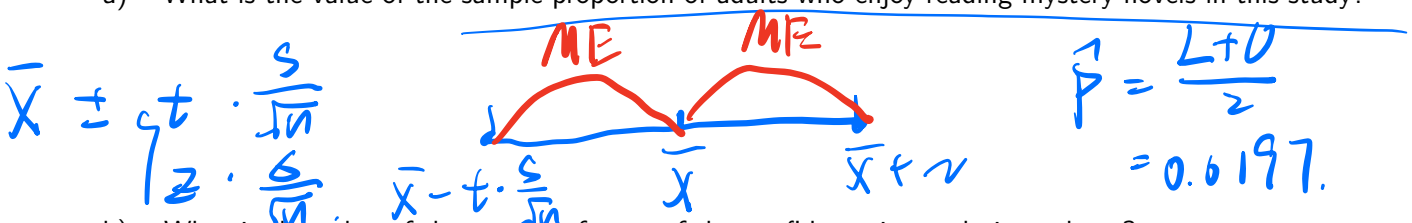
   we have sufficient evidence to

   Describe: support H_A / reject H_0

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2. The Society of Mystery Writers is interested in the proportion of adults who enjoy reading mystery novels. They conduct a study from which they find a 95% confidence interval for the population proportion to be (0.5891, 0.6503).

   a) What is the value of the sample proportion of adults who enjoy reading mystery novels in this study?



   b) What is the value of the margin of error of the confidence interval given above?

3. A recent study of 200 randomly selected children found that 27% like sour candies. Assume that you have a Binomial Experiment (i.e. you do not need to check the Binomial criteria).

   a) What is the appropriate distribution to use to construct a CI for p? (Check the relevant criteria.)

   b) Construct a 98% confidence interval for the population proportion of children who like sour candies.

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